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Intrapartum assessment of fetal wellbeing

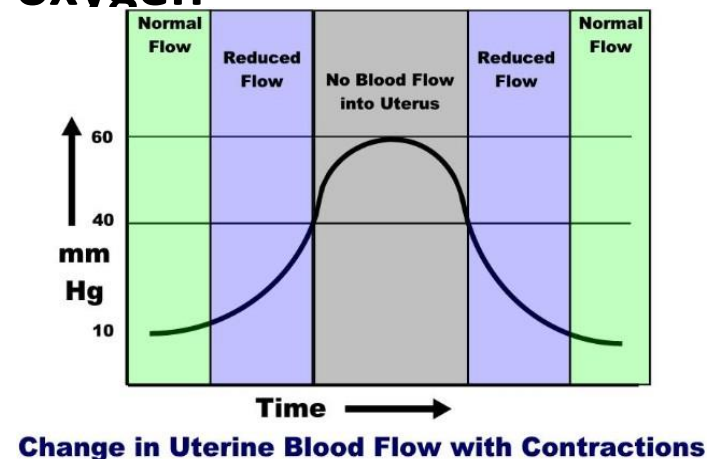
(Fetal monitoring in labour)

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Learning objectives:

- To gain knowledge about:
- Importance of fetal monitoring in labour.
- Causes of fetal hypoxia in labour.
- Response of the fetus to hypoxia in labour.
- Methods of monitoring fetal wellbeing in labour.
- What is normal and abnormal fetal heart rate in labour.
- How to manage abnormal fetal heart rate in labour.

- Fetal surveillance during labour is an essential element of good obstetric care because of the fact that Intrapartum hypoxia and acidosis may develop in any labour even with normal healthy pregnancy.
- With each contraction, placental blood flow and oxygen transfer to the fetus are temporarily interrupted.
- A healthy term fetus can usually withstand the stress associated with uterine contractions without suffering from hypoxia because sufficient oxygen exchange occurs during the interval between contractions.
- While a fetus whose oxygen supply is marginal cannot tolerate the stress of contractions and will become hypoxic .



Fetal physiology



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- Normal utero-placental unit:
- The fetus is sustained by the utero-placental unit and under normal circumstances provision is surplus to requirements, enabling the fetus to thrive, grow and build up energy reserves.
- Aerobic metabolism of glucose provides the fetus with its energy, and the oxygen which is vital for this process is usually plentiful.
- In normal pregnancy there is therefore more oxygen supplied by the utero-placental unit than needed by the fetus and energy reserves in the form of glycogen are stored by the fetus.

Fetal compromise :



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- Hypoxia during labour could be:
- 1. Acute hypoxia.
- If oxygen delivery declined , there is reduced oxygen in the fetal circulation (called hypoxemia).
- if levels fall further, the oxygen concentration in the fetal tissues reduced (hypoxia).
- Two things occur in response to hypoxemia and hypoxia:
- 1. The circulation adapts to maintain supplies to vital organs (brain and heart).
- 2. Anaerobic metabolism will start utilizing glucose released from the stored glycogen reserves provides continued energy supplies.
- The main consequences of anaerobic metabolism, which is almost 20 times less efficient than aerobic metabolism, is the accumulation of lactic acid, which is the waste product of this metabolism, results in a metabolic acidosis that can be harmful to fetal tissues.

Acute fetal hypoxia may happen suddenly during labour as a consequence of:

-Uterine hyper-stimulation, Placental abruption, or cord compression.



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2.Chronic hypoxia due to utero-placental perfusion insufficiency (during pregnancy):

-As in IUGR & PET, here there is impaired placental perfusion which develops during pregnancy leads to gradual deterioration in oxygen and nutrients supply to the fetus.

-The fetal circulation adapts as mentioned above with shifting blood to vital organs (brain and heart) and the peripheral circulation is effectively reduced.

-Depending on the severity and duration of this compromise, the pathophysiology results in decreased renal perfusion which in turn produces fetal oliguria (oligohydramnios) and reduced fetal growth and reduced fetal activity (decreased fetal movement).

-If there is further deterioration, anaerobic metabolism occurs and a metabolic acidosis develops.

-Acute on chronic.

-If an acute insult occurs in a fetus already compromised, it can withstand hypoxia less than a well-nourished fetus, so deterioration and hypoxia will develop rapidly during labour.

Fetal heart rate changes in relation to hypoxia:

- Under normal circumstances, the FHR is determined by the atrial pacemaker.**
- Modulation of the rate occurs physiologically through innervations of the heart by the parasympathetic nerves (decelerator) and the sympathetic nerves. (accelerator)**

- The initial response to chronic or slowly developing hypoxia is to increase cardiac output and re-distribute blood to the brain and heart.
- The increase in cardiac output is achieved by an increase in heart rate (tachycardia).
- This may be followed by a reduction in heart-rate variability due to brain stem hypoxia.
- Continued and worsening hypoxia will eventually leads to bradycardia & decelerations due to myocardial ischemia.
- While in acute hypoxia there is no time for fetal adaptation and it cause bradycardia & decelerations that is initially produced by chemoreceptor-mediated vagal stimulation and later by myocardial ischemia.

- Prolonged hypoxia leads to switch from aerobic to anaerobic metabolism and results in the generation of lactic acid and hydrogen ions.
- In excess, these saturate the buffering systems of the fetus and cause a metabolic acidosis, which if prolonged and severe, can cause neuronal damage and permanent neurological injury, even intrapartum fetal death.
- Hypoxia and acidosis cause a characteristic change in the fetal heart rate (FHR) pattern, which can be detected by auscultation and by the CTG.
- Clinical and experimental data indicate that fetal death occurs when 50% or more of the trans-placental oxygen exchange is interrupted.

Methods used to assess fetal wellbeing in labour:



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1. Monitoring the Fetal heart rate: this is the main method of monitoring the fetus during labour.

2. Assessing the amount and colour of amniotic fluid (after rupture of the membranes):

- Presence of any of the following are markers of potential fetal compromise.
- Absence of fluid (good liquor volume is a reassuring sign that the fetus has not been subjected to chronic hypoxia in the antenatal period).
- Meconium stained liquor.
- Heavy blood-stained fluid or bleeding.

3. Fetal scalp blood sampling (FBS): to measure the PH or lactate of the fetal blood.

Heart rate monitoring techniques:

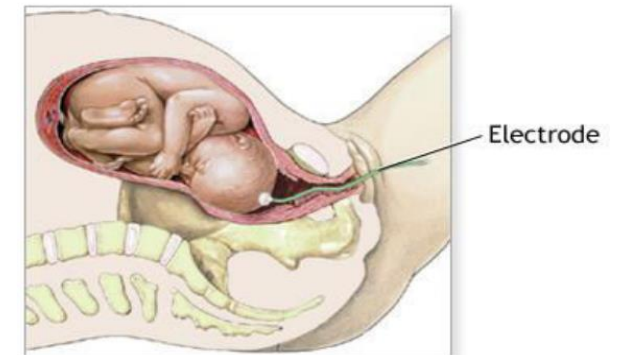
1. Intermittent auscultation of the fetal heart using a Pinard stethoscope or a hand-held Doppler ultrasound (sonicaid).



2. Continuous electronic fetal monitoring (EFM) using Cardiotocography (CTG):

A. Continuous external electronic fetal monitoring using external transducer.

B. Continuous internal electronic fetal monitoring using a fetal scalp electrode (FSE) and CTG.



Internal fetal monitoring

Intermittent auscultation of the fetal heart:

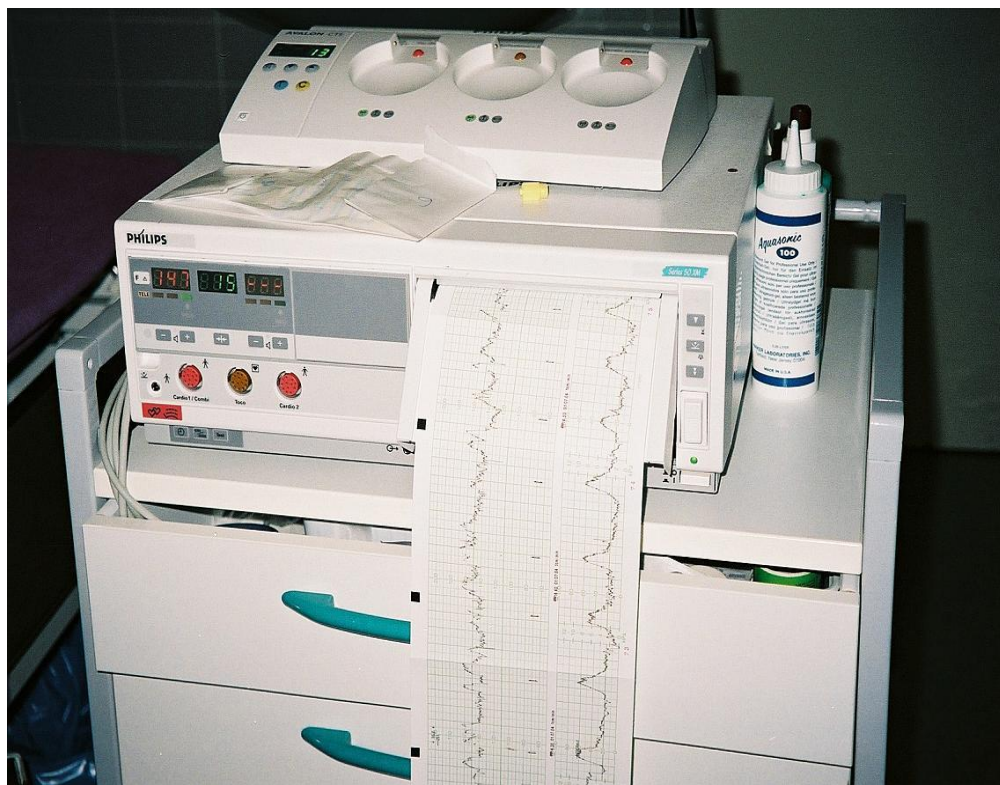


- This done by using a Pinard stethoscope or a hand-held Doppler ultrasound (sonicaid).
- This is for a woman with a normal uncomplicated pregnancy with a healthy fetus (low risk pregnancy).
- Frequency of auscultation:
- First stage of labour: - **latent phase** : every 30 min
 - **active phase** : every 15 min.
- 2nd stage of labour : every 5 min
or after each contraction.
- At which time we should listen and for how long?
- According to NICE listening should be after a contraction for 1 minute (60 seconds) on each occasion.
- FIGO recommends listening for at least 60 seconds 'during and for at least 30 seconds after a contraction'.
- This approach will detect the benign early decelerations of head compression commonly present in the late first stage and second stage of labour, and may miss late decelerations.

Continuous electronic fetal monitoring (EFM) using Cardiotocography (CTG).

- The cardiotocography (CTG) is a continuous tracing of the fetal heart rate used to assess fetal wellbeing, together with an assessment of uterine activity.
- The CTG recording is obtained with the pregnant woman positioned comfortably in a left lateral or semi-recumbent position to avoid compression of the maternal vena cava.
- Two external transducers are placed on the mother's abdomen, each attached with a belt.
- One transducer is a pressure-sensitive contraction toco-dyno-meter to measure uterine contractions
- The second transducer uses ultrasound and the Doppler effect to detect motion of the fetal heart, and measures the interval between successive beats, thereby allowing a continuous assessment of fetal heart rate.

Cardiotocography:



Indications for continuous EFM

1. Pregnant women with high risk factors (high risk pregnancy e.g.: pre-eclampsia, IUGR, prolonged pregnancy, GDM, Rh isoimmunization, APH...ect).

2. Abnormal FHR detected by intermittent auscultation (tachycardia, bradycardia, and decelerations).

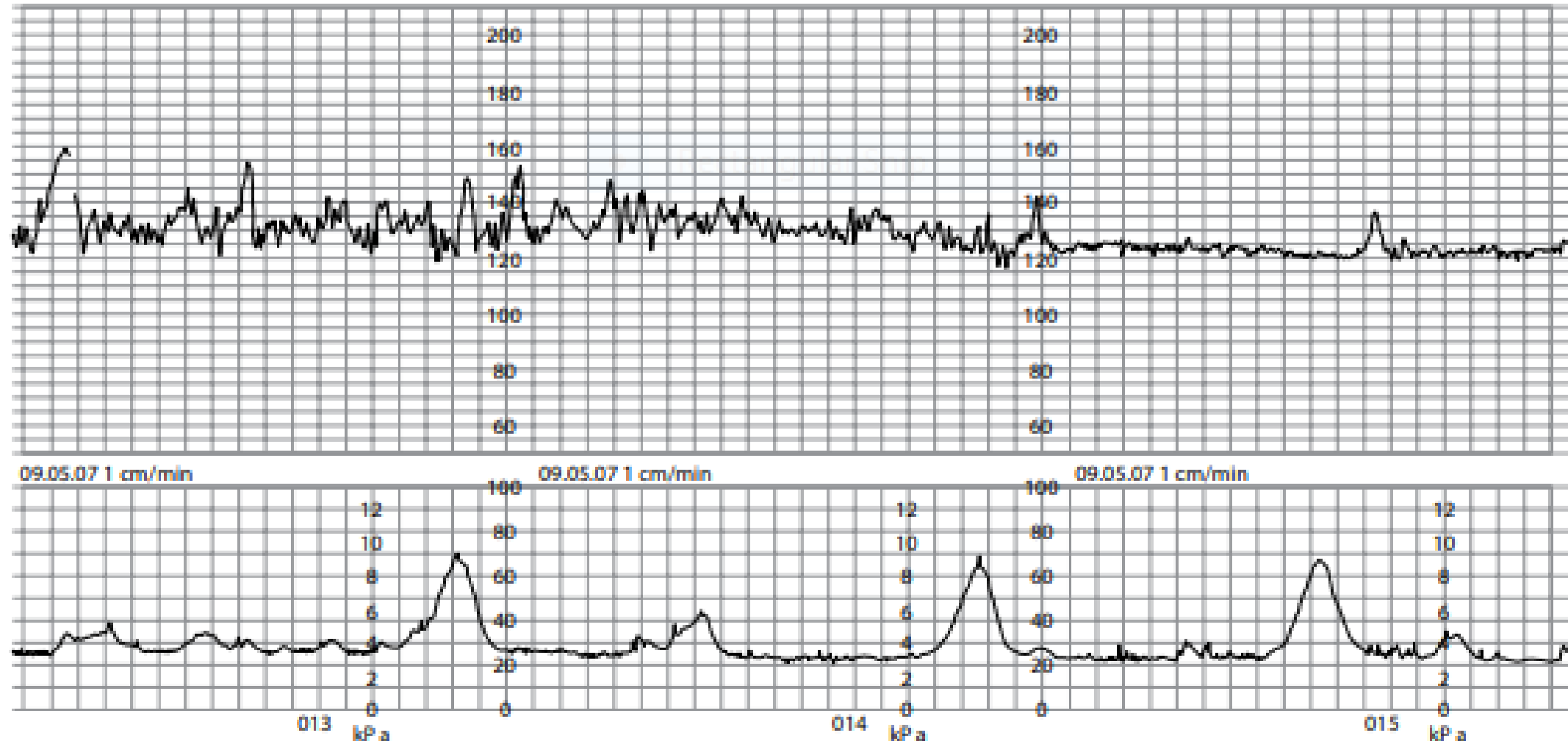
3. Events that arise during labour e.g:

- Significant meconium staining of the amniotic fluid.
- Fresh vaginal bleeding.
- Augmentation of contractions with oxytocin infusion.

Features of a normal FHR pattern on CTG include:

1. A baseline heart rate of between 110 -160 bpm. (for 20 minute interval or more).
 - The baseline fetal heart rate falls with advancing gestational age as a result of maturing fetal parasympathetic tone.
 - Before term 160 bpm is taken as upper limit of normal.
2. Baseline variability: (variation in the FHR above and below the baseline),
 - It is due to the balance between the parasympathetic and the sympathetic nervous system.
 - Normal baseline variability: 5-25 bpm.
3. Presence of accelerations (a transient increase in FHR of at least 15 bpm lasting at least 15 seconds).
4. Absence of decelerations (transient decrease in the FHR of 15 bpm or more lasting at least 15seconds).

A normal cardiotocograph (CTG), showing a baseline fetal heart rate of approximately 120 bpm, frequent accelerations, baseline variability of 10–15 bpm and no decelerations. The uterus is contracting approximately once every 5 minutes.



Abnormal Fetal heart rate patterns by CTG:

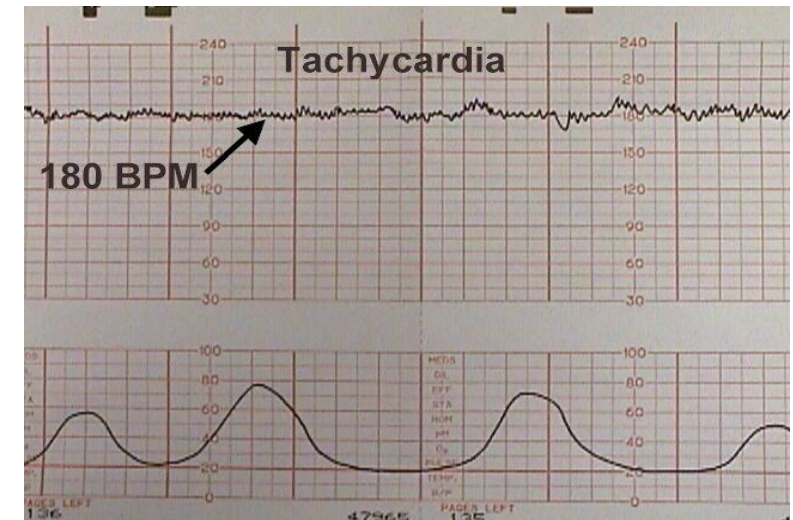
Abnormalities in baseline rate:

1.Fetal tachycardia: baseline rate > 160 bpm persistent for more than 10min.

- Moderate tachycardia (161-180) is a non-reassuring sign on CTG.
- Sever tachycardia (> 180) is a pathological sign on CTG.

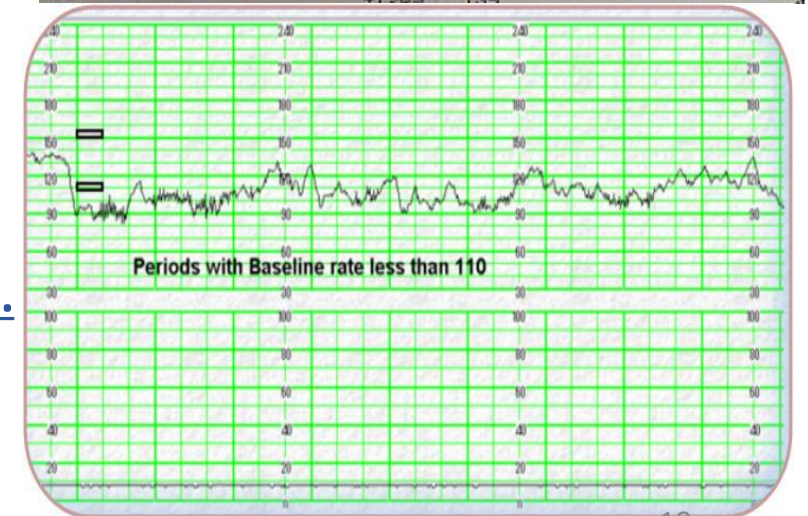
Tachycardia not always indicate fetal hypoxia and can be due to:

- Congenital tachycardia.
- Maternal tachycardia due to pain or dehydration.
- Maternal pyrexia or fetal infection.
- Fetal anemia.
- Drugs e.g. adreno-ceptor agonists(ritodrine).
- Fetal hypoxia.



2.Fetal bradycardia: baseline rate < 110 bpm for more than 10min.

- Moderate bradycardia (FHR : 100-109) is a non-reassuring sign on CTG.
- Persistent sever bradycardia (<100 bpm) is a pathological sign and usually associated with fetal hypoxia.



-Abnormalities in Baseline variability:

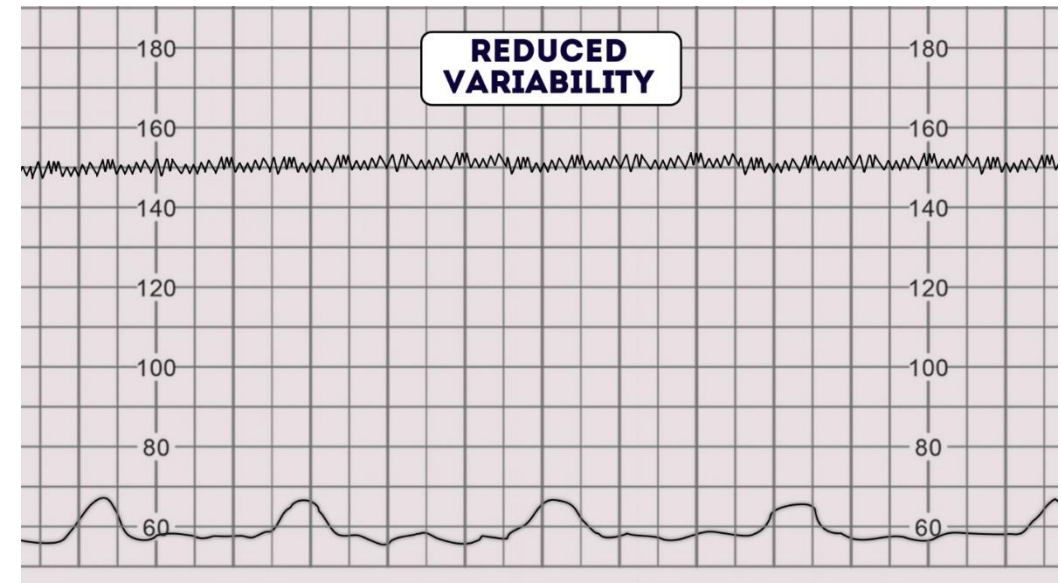
1. Reduced or Loss of baseline variability (less than 5bpm). may be caused by:

- Administration of drugs to the mother including:

-pethidine, diazepam and many anti-hypertensive agents, especially B-blockers.

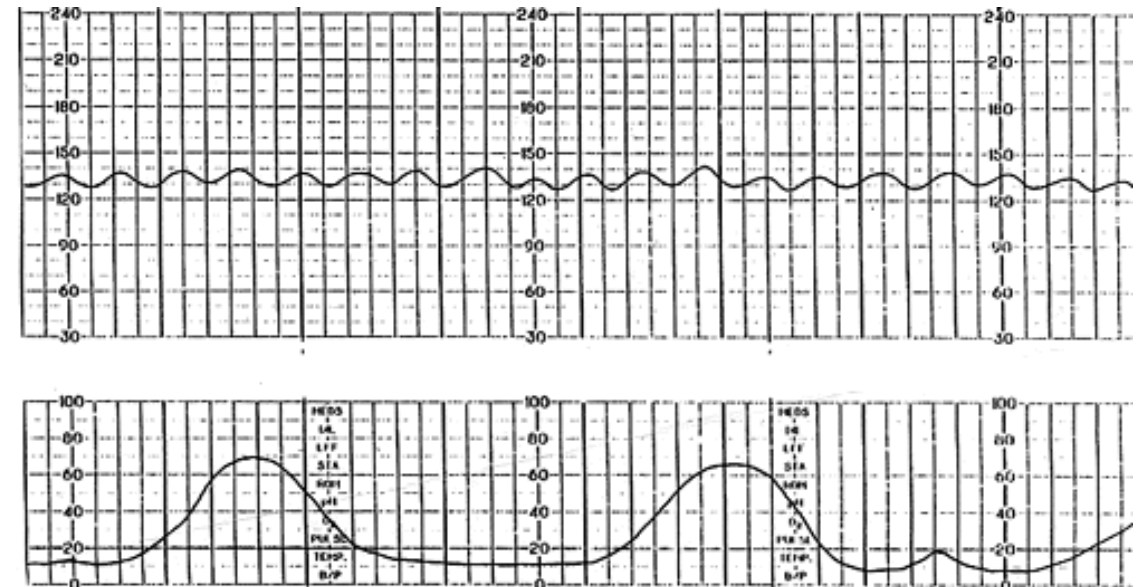
- Fetal sleep especially in early labour (usually for 30min.).

- Fetal hypoxia.



2. Increased baseline variability > 25bpm (Sinusoidal rhythm):

- It is a regular, smooth, undulating form typical of a sine wave that occurs with a frequency of 2-5 cycles per minute and an amplitude >25bpm.
- It is also characterized by a stable baseline heart rate of 120-160 bpm and absent beat-to-beat variability.
- The true sinusoidal pattern is rare but ominous and is associated with high rates of fetal morbidity and mortality.
- It indicates severe fetal anemia, as occurs in cases of
- Rh incompatibility or severe hypoxia.



Periodic changes in FHR

-These are changes in baseline FHR related to uterine contractions.

The responses to uterine contractions may be one of three:

1.No change in FHR during contractions.

2.Acceleration: The FHR increases in response to uterine contractions.

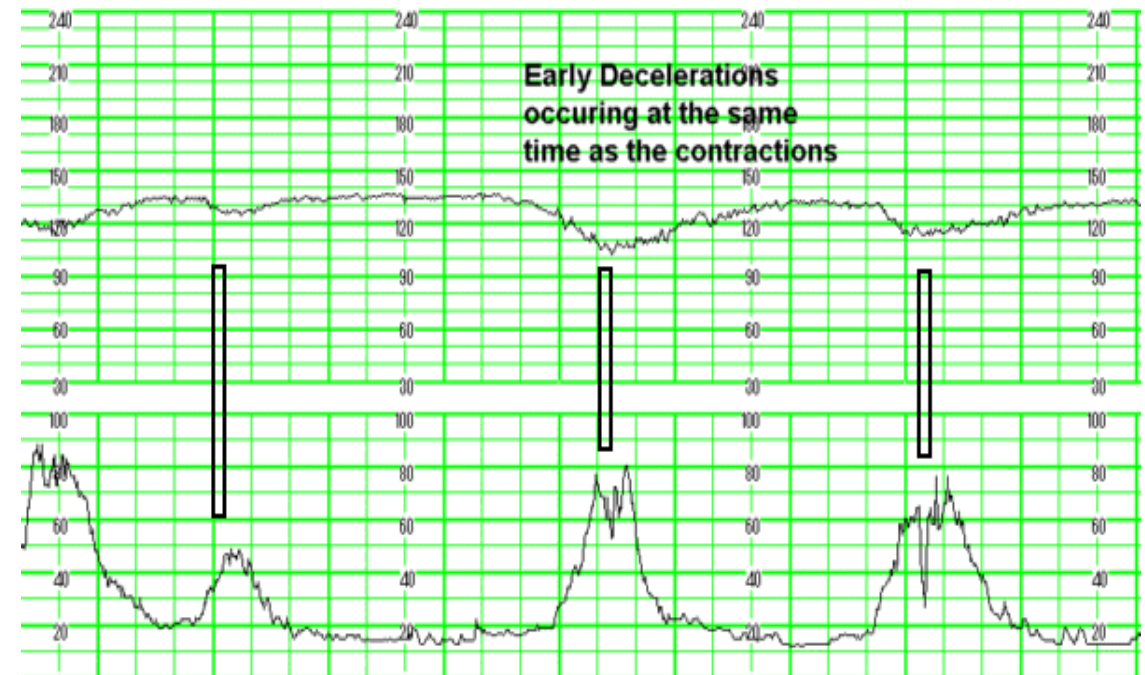
-Presence of accelerations is an excellent indicator of good fetal outcome.

3.Deceleration. The FHR decreases in response to uterine contractions.

- Decelerations may be early, late, variable, or mixed.
- All types of decelerations are abnormal except early (considered physiological).

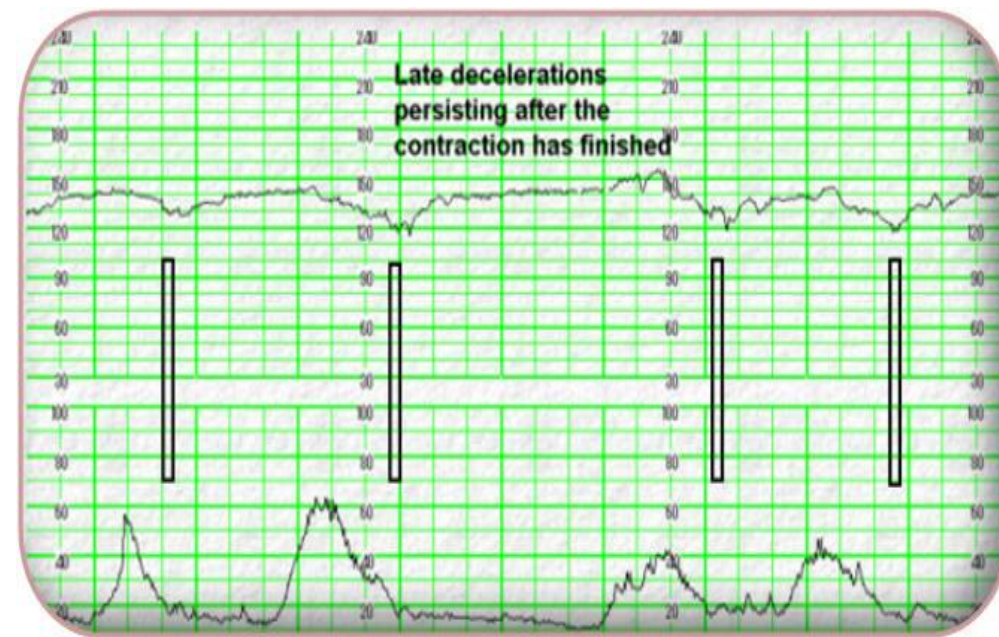
Early decelerations (head compression):

- This pattern usually has an onset, maximum fall, and recovery that is coincident with the onset, peak, and end of the uterine contraction.
- This pattern is seen when engagement of the fetal head has occurred (during late 1st stage or 2nd stage).
- The pressure on the fetal head leads to increased intracranial pressure that stimulate a vagal response leads to slowing of the FHR.
- Early decelerations are physiological & not associated with fetal hypoxia or low Apgar score.



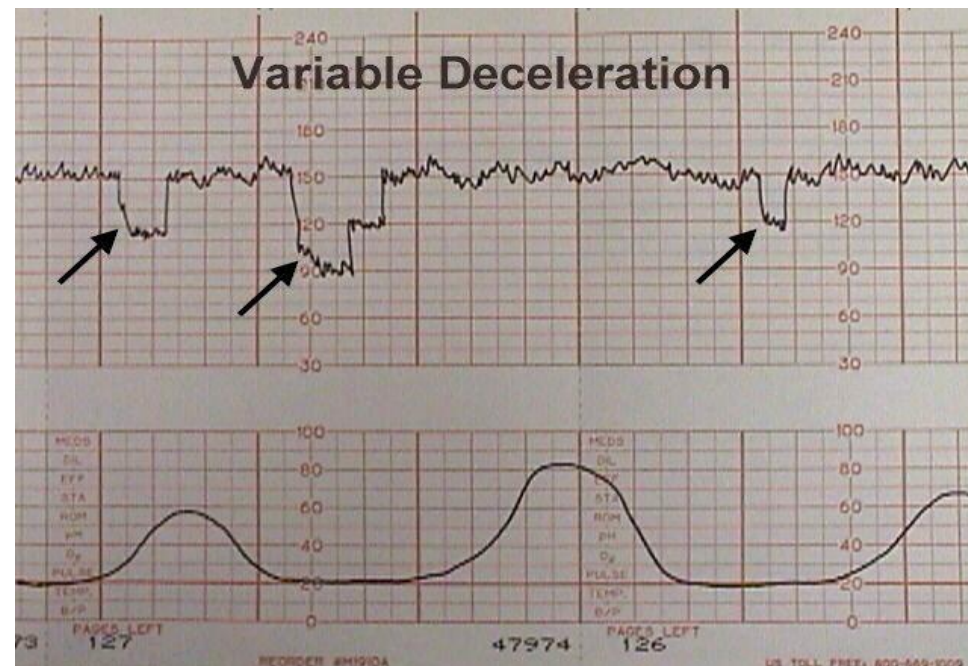
2 . Late decelerations (utero-placental insufficiency):

- Uniform, repetitive, periodic slowing of FHR with onset mid to end of the contraction and nadir more than 20 seconds after the peak of the contraction and ending after the contraction.
- This pattern has an onset, maximal decrease, and recovery that is shifted to the right in relation to the contraction.
- Multiple late decelerations are associated with placental insufficiency.
- Late decelerations are generally associated with low scalp blood pH values and high base deficits, indicating metabolic acidosis from anaerobic metabolism (fetal hypoxia).



Variable deceleration (cord compression).

- Variable, intermittent periodic slowing of FHR with rapid onset and recovery.
- Time relationships with contraction cycle are variable and they may occur in isolation.
- Variable decelerations usually indicate umbilical cord compression.

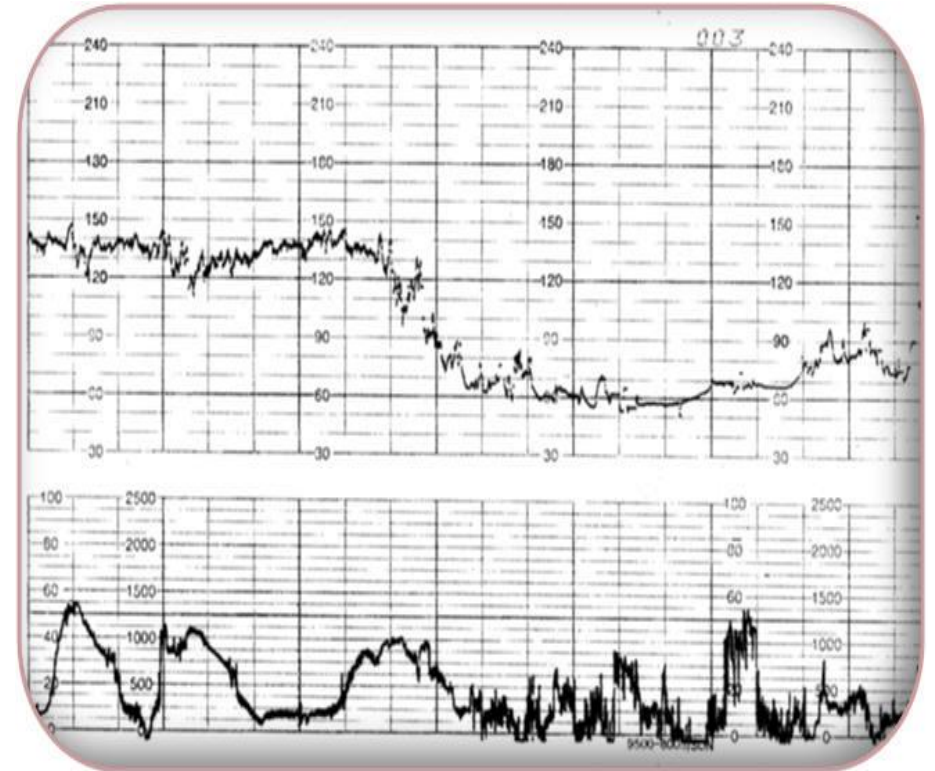


Prolonged deceleration:

An abrupt decrease in FHR to levels below the baseline that lasts at least 60–90 seconds.

-These decelerations become pathological if they cross two contractions, i.e. last > 3 minutes.

-There is an association between prolonged decelerations and fetal acidosis.



Interpretation of GTC records:



1.Normal/Reassuring Trace:

- Baseline heart rate 110-160 bpm.
- Baseline variability 5-25 bpm.
- No decelerations(or just early decelerations).
- Presence of accelerations(however the significance of no accelerations on an otherwise normal CTG in labour is unclear).

2.Non- reassuring CTG record:

The presence of one non-reassuring features with normal other features

1.Baseline heart rate: moderate bradycardia (100–109bpm).

or moderate tachycardia (161–180bpm).

2. Baseline variability: reduced <5bpm for 30–50mins.

or increased >25bpm for 15–25 minutes.

3. Variable or late decelerations: lasting< 30minutes

3. Abnormal /pathological CTG trace:

-The presence of two non-reassuring features.

Or one of the following pathological features:

1. Baseline HR: Sever tachycardia (FHR >180),
Sever bradycardia (FHR <100).
2. Baseline variability: -reduced variability (< 5bpm) for >50minutes
- or Sinusoidal pattern.
3. Repetitive variable or late decelerations for > 30minutes.
4. Single prolonged decelerations for >3minutes.

Strategies for intervention

- Strategies for intervention always depend on the clinical circumstances.
- When abnormal FHR patterns are seen, the first step should be a search for the underlying cause.
- When the cause is identified, such as maternal hypotension, steps should be taken to correct the problem.
- In general, a term fetus tolerates ominous fetal heart patterns better than a preterm fetus.
- Other considerations in the management of fetal distress include the maternal condition and the stage of labour.

Management of abnormal CTG :



A. Normal/Reassuring Trace:

-Risk of fetal hypoxia in spontaneous labour is low (Manage normally).

B . Suspicious /non-reassuring Trace

- Continue EFM but amniotomy should be performed:
- If meconium stained liquor is present perform fetal scalp blood pH (or deliver by C/S if not available).
- If liquor is clear : continue EFM.

C. Abnormal /Pathological Trace

- 1- Consider fetal scalp blood pH measurement to determine subsequent management.
- 2 . Or deliver by C/S : if fetal scalp sampling cannot be performed .

Emergency Interventions for Non-reassuring Patterns:

- 1 . Call for assistance.**
- 2 . Administer oxygen through a tight-fitting face mask.**
- 3 . Change maternal position (left lateral) to ovoid maternal hypotension.**
- 4 . Administer fluid bolus (lactated Ringer's solution).**
- 5 . Perform a vaginal examination and fetal scalp stimulation.**
- 6 . Discontinue Oxytocin if used.**
- 7 . Consider tocolysis (for uterine hyper stimulation).**
- 8. Do rupture of membranes to assess the amniotic fluids:**
 - If fluid is clear continue monitoring**
 - if fluid is meconium stained, consider fetal scalp sampling to determine fetal PH, if less than 7.2 deliver by C/S**
 - If fetal scalp can't be done or the abnormality persist > 30min. Deliver by C/S.**

Meconium staining of amniotic fluid:

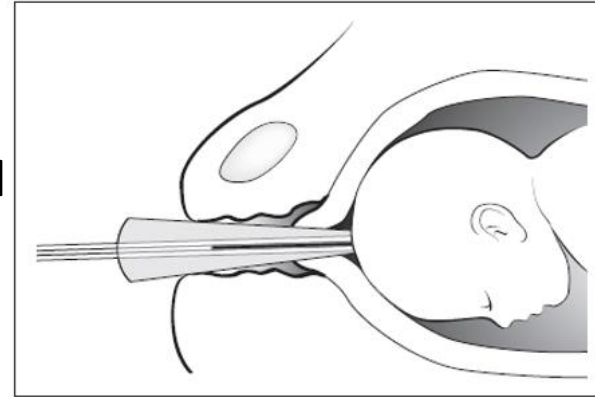


- A term fetus may have passed meconium by nature of its maturity.
 - If the fetus is preterm, the passage of meconium is not normal and may suggest infection or hypoxia.
- Meconium stained liquor is classified into 3 grades based on its colour and viscosity.
- 1-Light meconium (grade 1):** is lightly stained yellow or greenish amniotic fluid and the amount of fluid is good (Light passage is not associated with poor outcome).
- 2. Moderate meconium (grade 2):** dark green but still watery fluid.
- 3.Heavy meconium (grade 3):** is dark green or black and is usually thick and tenacious.
- If the meconium is thick, then by definition the liquor volume is reduced and reflects utero-placental insufficiency and possible fetal compromise.
- Heavy passage is associated with lower 1 and 5 minute Apgar scores** and is associated with the risk of meconium aspiration during delivery

- **Meconium** can be also classified into **early** and **late**.
- **Early passage** occurs any time **prior to rupture of the membranes**
- **Late passage** usually occurs **during the second stage of labour**, after clear amniotic fluid has been noted earlier.
- If the **liquor has been clear** and then **becomes meconium stained during labour**, it suggests that the fetus may be compromised and this could be due to either **hypoxia or infection**.

Fetal scalp blood sampling

-Fetal scalp blood sampling for pH determination is indicated when fetal distress is suggested by clinical parameters, such as heavy meconium, or by abnormal FHR patterns.



-Blood is obtained from the fetus by placing an amnio-scope trans-vaginally against the fetal skull.
The cx. Should be dilated at least 2cm, and the membranes should be ruptured

-A 2 × 2-mm lancet is used for a stab incision, and a drop of blood is aspirated into a long heparinized glass capillary tube.

-Normal pH ≥ 7.25 -PH= 7.2-7.25 (pre-asphaxia). PH ≤ 7.2 are abnormal (acidosis).

-The false-positive rate is about 8%, and the false-negative rate about 10%.

Complication: Infection and bleeding.

Fetal blood pH correctly predicts neonatal outcome in 82% of the time, as measured by the Apgar score.

A close-up photograph of a vibrant pink flower with a yellow center, set against a clear blue sky. The flower has many thin, pointed petals radiating from the center. The text "Thank you" is overlaid in a green, stylized font with a white outline.

Thank you